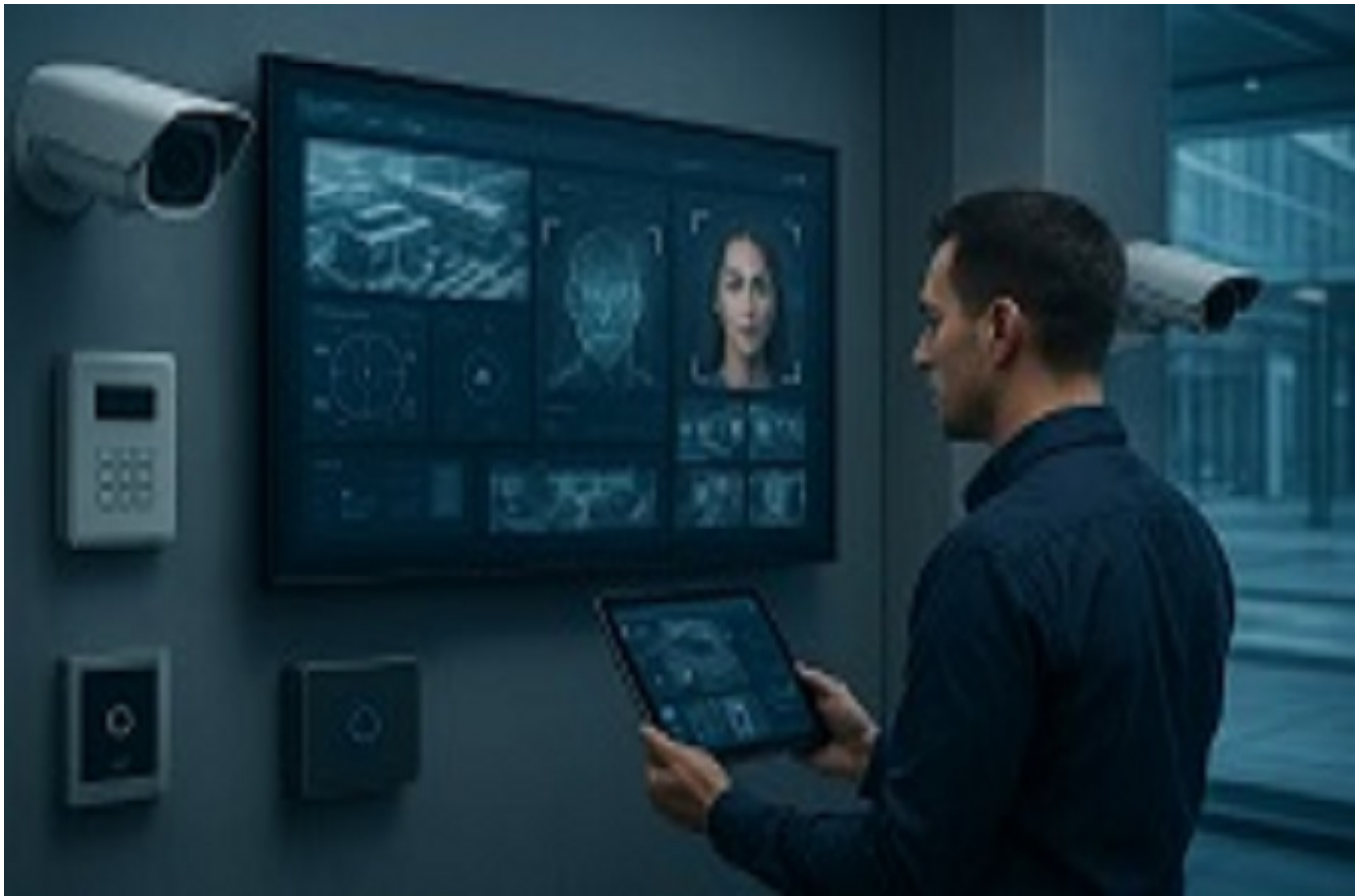


Qualification Pack



Intelligent Security Technology

QP Code: ELE/Q6705

Version: 1.0

NSQF Level: 4.5

Electronics Sector Skills Council of India || 155, 2nd Floor, ESC House Okhla Industrial Area-Phase 3
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Qualification Pack

ELE/Q6705: Intelligent Security Technology

Brief Job Description

This training program equips learners with the knowledge and skills required to design, install, optimize, and maintain modern security and intelligent systems. It combines engineering fundamentals, safety and regulatory compliance, digital and AI technologies, and practical industry practices to prepare professionals for end-to-end security system implementation and operations.

Personal Attributes

Detail-oriented, safety-conscious, and technically competent professional with strong analytical thinking, problem-solving ability, teamwork, and commitment to continuous learning in intelligent security technologies.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N6716: Work organization and management \(Intelligent Security Technology\)](#)
2. [ELE/N6717: Communication and interpersonal skills \(Intelligent Security Technology\)](#)
3. [ELE/N6724: Project initiation and holistic planning. \(Intelligent Security Technology\)](#)
4. [ELE/N6720: Planning and design of security systems \(Intelligent Security Technology\)](#)
5. [ELE/N6719: Installation and implementation of security systems \(Intelligent Security Technology\)](#)
6. [ELE/N6721: Debugging and optimization of security systems \(Intelligent Security Technology\)](#)
7. [ELE/N6722: Application of artificial intelligence and digital technologies \(Intelligent Security Technology\)](#)
8. [ELE/N6723: Maintenance and repair](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Security Surveillance
Occupation	Product Design & Development-EM&B

Qualification Pack

Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7411.0102
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-4.5-EH-05199-2026-V1-ESSCI
NQR Version	1

Remarks:

NA

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ELE/N6716: Work organization and management (Intelligent Security Technology)

Description

This NOS is about to plan, organize, and manage tasks effectively to ensure timely completion of work while maintaining quality and safety standards. Coordinate resources, communicate efficiently with team members, and demonstrate responsibility, discipline, and professional work ethics.

Scope

The scope covers the following :

- Work organization and management

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Comply with applicable health, safety, and environmental regulations, including correct selection and use of personal protective equipment (PPE), to ensure safe working practices at all times.
- PC2.** Apply first aid principles and emergency response procedures effectively to address workplace incidents and minimize risk to personnel and assets.
- PC3.** Interpret and adhere to international, national, and organizational standards, regulations, and guidelines relevant to the construction and implementation of security and protection systems.
- PC4.** Follow official approval and compliance processes required for constructing security and protection systems, in accordance with system type, purpose, and jurisdictional requirements.
- PC5.** Create, read, and interpret engineering drawings, layout plans, and configuration documents to ensure accurate installation and integration of security and protection systems.
- PC6.** Plan and manage project activities by applying principles of cost control, quality assurance, scheduling, and timely completion to achieve optimal project efficiency.
- PC7.** Implement electrical safety measures, including grounding, bonding, and lightning protection practices, during construction and installation activities.
- PC8.** Select, use, maintain, clean, and store tools, equipment, and materials safely and efficiently, while managing inventory, material flow, and on-site logistics.
- PC9.** Maintain accurate records of work progress, design changes, inspections, and test results, ensuring data integrity, security, and traceability for operation and maintenance purposes.
- PC10.** Demonstrate professional development by staying updated with emerging technologies, regulations, systems, and methods, and proactively sharing knowledge and improvement ideas with stakeholders.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Knowledge of applicable health, safety, and environmental regulations, including proper selection and use of PPE.
- KU2.** Understanding of first aid principles, emergency response procedures, and risk mitigation practices in the workplace.
- KU3.** Knowledge of international, national, and organizational standards related to security and protection system construction and implementation
- KU4.** Understanding of engineering drawings, layout plans, technical documentation, and system configuration requirements
- KU5.** Knowledge of project management principles, electrical safety practices (grounding, bonding, lightning protection), documentation processes, and inventory management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to apply safe working practices and respond effectively to emergencies and workplace incidents.
- GS2.** Skill in interpreting technical drawings, standards, and compliance requirements accurately.
- GS3.** Ability to plan, organize, and manage projects efficiently while maintaining quality and cost control.
- GS4.** Skill in handling, maintaining, and managing tools, equipment, materials, and site logistics responsibly.
- GS5.** Ability to maintain accurate documentation, ensure data integrity, and continuously update professional knowledge and competencies.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Comply with applicable health, safety, and environmental regulations, including correct selection and use of personal protective equipment (PPE), to ensure safe working practices at all times.	-	-	-	-
PC2. Apply first aid principles and emergency response procedures effectively to address workplace incidents and minimize risk to personnel and assets.	-	1	-	-
PC3. Interpret and adhere to international, national, and organizational standards, regulations, and guidelines relevant to the construction and implementation of security and protection systems.	-	-	-	-
PC4. Follow official approval and compliance processes required for constructing security and protection systems, in accordance with system type, purpose, and jurisdictional requirements.	-	1	-	-
PC5. Create, read, and interpret engineering drawings, layout plans, and configuration documents to ensure accurate installation and integration of security and protection systems.	-	-	-	-
PC6. Plan and manage project activities by applying principles of cost control, quality assurance, scheduling, and timely completion to achieve optimal project efficiency.	-	1	-	-
PC7. Implement electrical safety measures, including grounding, bonding, and lightning protection practices, during construction and installation activities.	-	-	-	-
PC8. Select, use, maintain, clean, and store tools, equipment, and materials safely and efficiently, while managing inventory, material flow, and on-site logistics.	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Maintain accurate records of work progress, design changes, inspections, and test results, ensuring data integrity, security, and traceability for operation and maintenance purposes.	-	-	-	-
PC10. Demonstrate professional development by staying updated with emerging technologies, regulations, systems, and methods, and proactively sharing knowledge and improvement ideas with stakeholders.	-	1	-	-
NOS Total	-	5	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6716
NOS Name	Work organization and management (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6717: Communication and interpersonal skills (Intelligent Security Technology)

Description

This NOS is about to effectively communicate technical and non-technical information with clients, team members, and stakeholders to ensure clear understanding of security system requirements and operations. Demonstrate professionalism, teamwork, active listening, and problem-solving skills to build strong working relationships and ensure successful project execution in Intelligent Security Technology.

Scope

The scope covers the following :

- Communication and interpersonal skills

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Listen actively to colleagues and stakeholders, asking relevant questions to enhance understanding of complex situations and work requirements.
- PC2.** Communicate clearly and professionally with colleagues using appropriate verbal and written communication methods.
- PC3.** Adapt communication style and approach to meet the changing needs, roles, and expectations of colleagues and team members.
- PC4.** Establish and maintain effective working relationships with colleagues and managers to support collaborative and productive work environments.
- PC5.** Apply principles and techniques of teamwork to lead and/or support the development of strong, efficient, and cohesive teams.
- PC6.** Communicate effectively with external organizations, authorities, and stakeholders to obtain required project approvals, permissions, or consents.
- PC7.** Share professional knowledge and skills with colleagues and provide guidance, coaching, or support to enhance team capability and performance.
- PC8.** Identify, manage, and resolve misunderstandings, tension, and conflicts professionally to support positive and constructive outcomes.
- PC9.** Apply structured problem-solving processes and techniques to generate options, evaluate alternatives, and implement appropriate solutions.
- PC10.** Maintain concentration, focus, and professional demeanour while working in complex, multi-system, and high-pressure environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Listen actively to colleagues and stakeholders, asking relevant questions to enhance understanding of complex situations and work requirements.	-	-	-	-
PC2. Communicate clearly and professionally with colleagues using appropriate verbal and written communication methods.	-	1	-	-
PC3. Adapt communication style and approach to meet the changing needs, roles, and expectations of colleagues and team members.	-	-	-	-
PC4. Establish and maintain effective working relationships with colleagues and managers to support collaborative and productive work environments.	-	1	-	-
PC5. Apply principles and techniques of teamwork to lead and/or support the development of strong, efficient, and cohesive teams.	-	-	-	-
PC6. Communicate effectively with external organizations, authorities, and stakeholders to obtain required project approvals, permissions, or consents.	-	1	-	-
PC7. Share professional knowledge and skills with colleagues and provide guidance, coaching, or support to enhance team capability and performance.	-	-	-	-
PC8. Identify, manage, and resolve misunderstandings, tension, and conflicts professionally to support positive and constructive outcomes.	-	1	-	-
PC9. Apply structured problem-solving processes and techniques to generate options, evaluate alternatives, and implement appropriate solutions.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Maintain concentration, focus, and professional demeanour while working in complex, multi-system, and high-pressure environments.	-	1	-	-
NOS Total	-	5	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6717
NOS Name	Communication and interpersonal skills (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6724: Project initiation and holistic planning. (Intelligent Security Technology)

Description

This NOS is about to initiate security technology projects by analyzing client requirements, conducting site assessments, and defining scope, objectives, and technical specifications. Develop comprehensive project plans covering resources, timelines, risk management, compliance, and system integration to ensure efficient and successful implementation.

Scope

The scope covers the following :

- Project initiation and holistic planning

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare for participation in the sales process by reviewing organizational sales structures, procedures, market research, and consulting with relevant colleagues.
- PC2.** Identify and understand the range of roles and responsibilities that contribute to the sales and solution development process.
- PC3.** Engage with clients using structured questioning and analysis techniques to accurately define, interpret, and document client needs and expectations.
- PC4.** Operate within the scope and limits of one's professional role while engaging with clients, escalating requirements appropriately where necessary.
- PC5.** Inspect the asset requiring security solutions (such as a building, infrastructure, or environment) to validate, refine, or modify initial client requirements.
- PC6.** Analyze technical, operational, and financial feasibility of potential security solutions, including alternative options and their implications.
- PC7.** Formulate and present viable solution options in coordination with internal experts, professionals, and external suppliers.
- PC8.** Apply standard or bespoke costing and pricing methods to estimate installation costs, system components, and alternative solutions with financial comparisons.
- PC9.** Create and/or interpret technical specifications and blueprints using appropriate tools and software for security system infrastructure and components.
- PC10.** Liaise with relevant colleagues and continuously review and validate design, costing, and documentation throughout the solution development process to ensure accuracy and compliance.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Prepare for participation in the sales process by reviewing organizational sales structures, procedures, market research, and consulting with relevant colleagues.	-	-	-	-
PC2. Identify and understand the range of roles and responsibilities that contribute to the sales and solution development process.	-	1	-	-
PC3. Engage with clients using structured questioning and analysis techniques to accurately define, interpret, and document client needs and expectations.	-	-	-	-
PC4. Operate within the scope and limits of one's professional role while engaging with clients, escalating requirements appropriately where necessary.	-	1	-	-
PC5. Inspect the asset requiring security solutions (such as a building, infrastructure, or environment) to validate, refine, or modify initial client requirements.	-	-	-	-
PC6. Analyze technical, operational, and financial feasibility of potential security solutions, including alternative options and their implications.	-	1	-	-
PC7. Formulate and present viable solution options in coordination with internal experts, professionals, and external suppliers.	-	-	-	-
PC8. Apply standard or bespoke costing and pricing methods to estimate installation costs, system components, and alternative solutions with financial comparisons.	-	1	-	-
PC9. Create and/or interpret technical specifications and blueprints using appropriate tools and software for security system infrastructure and components.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Liaise with relevant colleagues and continuously review and validate design, costing, and documentation throughout the solution development process to ensure accuracy and compliance.	-	1	-	-
NOS Total	-	5	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6724
NOS Name	Project initiation and holistic planning. (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6720: Planning and design of security systems (Intelligent Security Technology)

Description

This NOS is about to design and plan intelligent security systems by analyzing risk assessments, site conditions, and client requirements to develop technically sound and scalable solutions. Prepare system architecture, equipment selection, layouts, and integration plans to ensure optimal performance, compliance, and reliability.

Scope

The scope covers the following :

- Planning and design of security systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Use appropriate communication methods and structured questioning techniques to understand and document user and stakeholder requirements for security systems.
- PC2.** Apply recognized planning and design methodologies to develop effective and reliable security system solutions.
- PC3.** Interpret and comply with applicable standards governing infrastructure, auxiliary components, and system integration.
- PC4.** Read, interpret, and analyze technical drawings and specifications in accordance with recognized industry standards and regulations.
- PC5.** Create clear and compliant security management system diagrams and architecture layouts using appropriate drawing tools.
- PC6.** Apply safety standards and risk mitigation principles during the planning and design phases of security system projects.
- PC7.** Design scientifically sound, practical, and scalable security system solutions aligned to specific technologies and products.
- PC8.** Evaluate project environments and site conditions to enhance system safety, reliability, and performance.
- PC9.** Analyze project plans and specifications for intelligent security systems to ensure technical feasibility and compliance.
- PC10.** Develop detailed design plans and documentation in compliance with applicable international standards and best practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Comprehensive knowledge of communication techniques, structured questioning methods, and stakeholder engagement practices for accurately capturing and documenting security system requirements.
- KU2.** Understanding of recognized planning and design methodologies, system engineering principles, and solution development processes applicable to intelligent security systems.
- KU3.** Knowledge of applicable national and international standards, codes, regulations, and compliance requirements related to infrastructure, auxiliary components, safety, and system integration.
- KU4.** Understanding of technical drawings, architectural layouts, schematics, and industry-standard symbols used in security system design and documentation.
- KU5.** Knowledge of safety standards, risk assessment principles, environmental evaluation methods, and best practices for developing reliable, scalable, and compliant security management system designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively communicate with users, clients, and stakeholders using structured questioning and documentation techniques to accurately capture system requirements.
- GS2.** Apply analytical and problem-solving skills to interpret technical drawings, project specifications, and site conditions to ensure system feasibility and compliance.
- GS3.** Utilize appropriate planning tools and drawing software to create clear, compliant, and professional security system diagrams and architecture layouts.
- GS4.** Evaluate risks, site environments, and operational constraints to design safe, reliable, and performance-optimized security system solutions.
- GS5.** Develop detailed, standards-compliant design documentation by integrating technical, regulatory, and project-specific requirements into practical and scalable solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	20	-	-
PC1. Use appropriate communication methods and structured questioning techniques to understand and document user and stakeholder requirements for security systems.	-	2	-	-
PC2. Apply recognized planning and design methodologies to develop effective and reliable security system solutions.	-	2	-	-
PC3. Interpret and comply with applicable standards governing infrastructure, auxiliary components, and system integration.	-	2	-	-
PC4. Read, interpret, and analyze technical drawings and specifications in accordance with recognized industry standards and regulations.	-	2	-	-
PC5. Create clear and compliant security management system diagrams and architecture layouts using appropriate drawing tools.	-	2	-	-
PC6. Apply safety standards and risk mitigation principles during the planning and design phases of security system projects.	-	2	-	-
PC7. Design scientifically sound, practical, and scalable security system solutions aligned to specific technologies and products.	-	2	-	-
PC8. Evaluate project environments and site conditions to enhance system safety, reliability, and performance.	-	2	-	-
PC9. Analyze project plans and specifications for intelligent security systems to ensure technical feasibility and compliance.	-	2	-	-
PC10. Develop detailed design plans and documentation in compliance with applicable international standards and best practices.	-	2	-	-
NOS Total	-	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6720
NOS Name	Planning and design of security systems (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6719: Installation and implementation of security systems (Intelligent Security Technology)

Description

This NOS is about to install and implement intelligent security systems by following technical specifications, safety standards, and approved design layouts. Ensure proper integration, configuration, testing, and commissioning of security equipment to achieve reliable and efficient system performance.

Scope

The scope covers the following :

- Installation and implementation of security systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analyze client requirements and application needs accurately to plan the installation and implementation of security system applications.
- PC2.** Evaluate construction and site environments to prepare appropriately for secure, efficient, and compliant project execution.
- PC3.** Select optimal security system design solutions that align with specific application requirements and operational conditions.
- PC4.** Read, interpret, and apply engineering drawings, wiring diagrams, schematics, technical manuals, and specifications accurately.
- PC5.** Assess system functionality and identify key operational and performance parameters for security system components.
- PC6.** Select appropriate components and devices based on technical specifications, application characteristics, and compatibility
- PC7.** Use specialized instruments, tools, and meters to support correct installation, testing, and verification of security systems.
- PC8.** Apply scientifically sound, energy-efficient, and environmentally sustainable installation practices.
- PC9.** Adhere strictly to applicable safety regulations and implement necessary safety measures throughout installation activities.
- PC10.** Complete security system installations accurately and systematically in accordance with approved construction drawings and specifications.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of client requirement analysis techniques and application assessment methods for effective planning and implementation of security system installations.

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- KU2.** Understanding of construction site conditions, environmental factors, infrastructure readiness, and regulatory compliance requirements related to security system projects.
- KU3.** Knowledge of engineering drawings, wiring diagrams, schematics, technical manuals, and relevant specifications used in security system installation and integration.
- KU4.** Understanding of security system components, devices, operational parameters, performance characteristics, and compatibility requirements.
- KU5.** Knowledge of installation methodologies, safety regulations, risk prevention measures, energy-efficient practices, and environmentally sustainable approaches applicable to security system projects.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze client requirements and site conditions to plan and execute secure, efficient, and compliant security system installations.
- GS2.** Interpret and apply technical drawings, schematics, manuals, and specifications accurately during installation and implementation.
- GS3.** Select appropriate components, tools, instruments, and testing equipment to ensure proper installation, functionality, and system verification.
- GS4.** Apply critical thinking and problem-solving skills to assess system performance, troubleshoot issues, and ensure operational efficiency.
- GS5.** Adhere to safety standards and implement systematic, environmentally responsible installation practices in alignment with approved designs and regulatory requirements.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Analyze client requirements and application needs accurately to plan the installation and implementation of security system applications.	-	1	-	-
PC2. Evaluate construction and site environments to prepare appropriately for secure, efficient, and compliant project execution.	-	1	-	-
PC3. Select optimal security system design solutions that align with specific application requirements and operational conditions.	-	1	-	-
PC4. Read, interpret, and apply engineering drawings, wiring diagrams, schematics, technical manuals, and specifications accurately.	-	1	-	-
PC5. Assess system functionality and identify key operational and performance parameters for security system components.	-	1	-	-
PC6. Select appropriate components and devices based on technical specifications, application characteristics, and compatibility	-	1	-	-
PC7. Use specialized instruments, tools, and meters to support correct installation, testing, and verification of security systems.	-	1	-	-
PC8. Apply scientifically sound, energy-efficient, and environmentally sustainable installation practices.	-	1	-	-
PC9. Adhere strictly to applicable safety regulations and implement necessary safety measures throughout installation activities.	-	1	-	-
PC10. Complete security system installations accurately and systematically in accordance with approved construction drawings and specifications.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6719
NOS Name	Installation and implementation of security systems (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6721: Debugging and optimization of security systems (Intelligent Security Technology)

Description

This NOS is about to Identify, troubleshoot, and rectify faults in intelligent security systems using systematic diagnostic methods and technical tools. Optimize system performance through configuration adjustments, firmware updates, and performance testing to ensure reliability and efficiency.

Scope

The scope covers the following :

- Debugging and optimization of security systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply fundamental theories of security system architecture to understand system structure, subsystem interactions, and operational flow.
- PC2.** Analyze connection methods, data flow paths, and communication protocols between security system subsystems.
- PC3.** Interpret and comply with applicable industry standards and specifications related to debugging, testing, and optimization of security systems.
- PC4.** Apply fundamental principles of debugging to identify faults, inefficiencies, and performance bottlenecks in security products.
- PC5.** Select appropriate application solutions for security systems based on functional, technical, and operational requirements.
- PC6.** Optimize security application systems to improve overall performance, reliability, and stability.
- PC7.** Apply principles of energy efficiency management to reduce power consumption while maintaining required system performance.
- PC8.** Enhance and optimize hardware performance of security products through calibration, tuning, and configuration adjustments.
- PC9.** Calibrate hardware devices and sensors within security systems to ensure accuracy, consistency, and reliable operation.
- PC10.** Utilize appropriate software debugging and diagnostic tools to commission, configure, and fine-tune security systems.
- PC11.** Debug communication, control, and integration issues across hardware and software interfaces within security systems.
- PC12.** Analyze and optimize system performance metrics and key indicators to meet design and operational benchmarks.
- PC13.** Apply control principles and mechanisms to improve responsiveness, stability, and automation of security application systems.

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- PC14.** Improve the usability and user experience of security products through performance tuning and functional optimization.
- PC15.** Document debugging, optimization actions, and performance improvements in accordance with organizational and industry requirements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of fundamental theories and principles of security system architecture, including system structure, subsystem integration, and operational workflows.
- KU2.** Understanding of connection methods, data transmission pathways, communication interfaces, and protocols used in integrated security systems.
- KU3.** Knowledge of industry standards, technical specifications, and compliance requirements related to system debugging, testing, commissioning, and optimization.
- KU4.** Understanding of debugging methodologies, fault detection techniques, performance analysis methods, and system optimization strategies for security products.
- KU5.** Knowledge of hardware calibration processes, energy efficiency management principles, software diagnostic tools, and performance enhancement techniques applicable to security systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze security system architecture, subsystem interactions, and communication flows to identify potential issues and improve system functionality.
- GS2.** Apply structured debugging and diagnostic techniques to detect faults, resolve inefficiencies, and eliminate performance bottlenecks.
- GS3.** Select and implement appropriate security application solutions based on functional, technical, and operational requirements.
- GS4.** Optimize hardware and software configurations through calibration, tuning, and energy-efficient adjustments to enhance reliability and stability.
- GS5.** Utilize appropriate software tools and testing instruments to commission, configure, fine-tune, and validate security system performance in compliance with applicable standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	30	-	-
PC1. Apply fundamental theories of security system architecture to understand system structure, subsystem interactions, and operational flow.	-	2	-	-
PC2. Analyze connection methods, data flow paths, and communication protocols between security system subsystems.	-	2	-	-
PC3. Interpret and comply with applicable industry standards and specifications related to debugging, testing, and optimization of security systems.	-	2	-	-
PC4. Apply fundamental principles of debugging to identify faults, inefficiencies, and performance bottlenecks in security products.	-	2	-	-
PC5. Select appropriate application solutions for security systems based on functional, technical, and operational requirements.	-	2	-	-
PC6. Optimize security application systems to improve overall performance, reliability, and stability.	-	2	-	-
PC7. Apply principles of energy efficiency management to reduce power consumption while maintaining required system performance.	-	2	-	-
PC8. Enhance and optimize hardware performance of security products through calibration, tuning, and configuration adjustments.	-	2	-	-
PC9. Calibrate hardware devices and sensors within security systems to ensure accuracy, consistency, and reliable operation.	-	2	-	-
PC10. Utilize appropriate software debugging and diagnostic tools to commission, configure, and fine-tune security systems.	-	2	-	-
PC11. Debug communication, control, and integration issues across hardware and software interfaces within security systems.	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Analyze and optimize system performance metrics and key indicators to meet design and operational benchmarks.	-	2	-	-
PC13. Apply control principles and mechanisms to improve responsiveness, stability, and automation of security application systems.	-	2	-	-
PC14. Improve the usability and user experience of security products through performance tuning and functional optimization.	-	2	-	-
PC15. Document debugging, optimization actions, and performance improvements in accordance with organizational and industry requirements.	-	2	-	-
NOS Total	-	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6721
NOS Name	Debugging and optimization of security systems (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6722: Application of artificial intelligence and digital technologies (Intelligent Security Technology)

Description

This NOS is about to apply artificial intelligence, analytics, and digital technologies to enhance the performance, automation, and predictive capabilities of intelligent security systems. Integrate smart algorithms, IoT platforms, and data-driven solutions to improve threat detection, monitoring accuracy, and decision-making efficiency.

Scope

The scope covers the following :

- Application of artificial intelligence and digital technologies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply fundamental theories of artificial intelligence to understand the capabilities and limitations of AI technologies in security and digital systems.
- PC2.** Identify and analyze application scenarios where digital and intelligent technologies can be effectively deployed.
- PC3.** Apply basic artificial intelligence algorithms to solve defined problems within digital and security applications.
- PC4.** Utilize foundational digital technology concepts to support the integration and operation of intelligent systems.
- PC5.** Integrate AI-related data sources, devices, and systems to enable intelligent functionality and data-driven decision-making.
- PC6.** Process data and adapt AI algorithms to meet specific application, performance, and operational requirements.
- PC7.** Design and construct intelligent security network architectures that support scalability, reliability, and interoperability.
- PC8.** Apply principles of cloud computing to support centralized processing, storage, and system management.
- PC9.** Coordinate and utilize edge computing and cloud computing collaboratively to optimize latency, bandwidth, and system performance.
- PC10.** Validate integrated AI and digital technology solutions to ensure functional, technical, and operational effectiveness.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Knowledge of fundamental theories of artificial intelligence, including machine learning concepts, data-driven modeling, algorithmic logic, and the capabilities and limitations of AI technologies in security and digital systems.
- KU2.** Understanding of digital technology foundations such as data structures, networking, IoT integration, intelligent devices, and system interoperability within security applications.
- KU3.** Knowledge of AI algorithms, data processing techniques, model adaptation methods, and performance optimization strategies for real-world applications.
- KU4.** Understanding of intelligent security network architecture design principles, including scalability, reliability, interoperability, and system integration frameworks.
- KU5.** Knowledge of cloud computing, edge computing, distributed processing models, centralized storage systems, and validation methodologies for integrated AI-driven solutions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze application scenarios to identify opportunities for deploying AI and digital technologies effectively within security and intelligent systems.
- GS2.** Apply appropriate AI algorithms and data processing techniques to solve defined technical problems and meet operational requirements.
- GS3.** Design and integrate intelligent network architectures by coordinating AI models, devices, data sources, and digital platforms.
- GS4.** Utilize cloud and edge computing collaboratively to optimize system latency, bandwidth usage, scalability, and overall performance.
- GS5.** Validate and evaluate integrated AI and digital solutions through systematic testing, performance monitoring, and compliance verification to ensure operational effectiveness.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Apply fundamental theories of artificial intelligence to understand the capabilities and limitations of AI technologies in security and digital systems.	-	1	-	-
PC2. Identify and analyze application scenarios where digital and intelligent technologies can be effectively deployed.	-	1	-	-
PC3. Apply basic artificial intelligence algorithms to solve defined problems within digital and security applications.	-	1	-	-
PC4. Utilize foundational digital technology concepts to support the integration and operation of intelligent systems.	-	1	-	-
PC5. Integrate AI-related data sources, devices, and systems to enable intelligent functionality and data-driven decision-making.	-	1	-	-
PC6. Process data and adapt AI algorithms to meet specific application, performance, and operational requirements.	-	1	-	-
PC7. Design and construct intelligent security network architectures that support scalability, reliability, and interoperability.	-	1	-	-
PC8. Apply principles of cloud computing to support centralized processing, storage, and system management.	-	1	-	-
PC9. Coordinate and utilize edge computing and cloud computing collaboratively to optimize latency, bandwidth, and system performance.	-	1	-	-
PC10. Validate integrated AI and digital technology solutions to ensure functional, technical, and operational effectiveness.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6722
NOS Name	Application of artificial intelligence and digital technologies (Intelligent Security Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6723: Maintenance and repair

Description

This NOS is about to perform preventive and corrective maintenance of intelligent security systems to ensure continuous, reliable, and safe operation. Diagnose faults, repair or replace defective components, and update system configurations to maintain optimal performance and compliance standards.

Scope

The scope covers the following :

- Maintenance and repair (Intelligent Security Technology)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor changes in risk profiles and client needs to determine appropriate maintenance, servicing, or upgrade actions for security systems.
- PC2.** Perform routine maintenance and servicing activities in accordance with established security industry maintenance standards and procedures.
- PC3.** Conduct software unit testing and ICT troubleshooting to identify faults and performance issues within security systems.
- PC4.** Test sub-systems and integrated components to verify correct operation or diagnose malfunctions accurately.
- PC5.** Inspect systems to identify problems such as loose or damaged connections, component degradation, or equipment failure.
- PC6.** Evaluate repair versus replacement options and carry out corrective actions to restore system functionality effectively.
- PC7.** Verify that existing security systems continue to meet current technical, safety, and regulatory standards.
- PC8.** Advise clients on equipment upgrades, regulatory changes, and functional improvements to maintain system effectiveness and client confidence.
- PC9.** Implement sustainable practices by recycling replaced equipment and materials in accordance with environmental guidelines.
- PC10.** Provide backup, disaster recovery support, and coordinate multi-system operational maintenance while completing and sharing required reports and documentation with relevant stakeholders.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of fundamental theories of artificial intelligence, including machine learning concepts, data-driven modeling, algorithmic logic, and the capabilities and limitations of AI technologies in security and digital systems.

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- KU2.** Understanding of digital technology foundations such as data structures, networking, IoT integration, intelligent devices, and system interoperability within security applications.
- KU3.** Knowledge of AI algorithms, data processing techniques, model adaptation methods, and performance optimization strategies for real-world applications.
- KU4.** Understanding of intelligent security network architecture design principles, including scalability, reliability, interoperability, and system integration frameworks.
- KU5.** Knowledge of cloud computing, edge computing, distributed processing models, centralized storage systems, and validation methodologies for integrated AI-driven solutions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze application scenarios to identify opportunities for deploying AI and digital technologies effectively within security and intelligent systems.
- GS2.** Apply appropriate AI algorithms and data processing techniques to solve defined technical problems and meet operational requirements.
- GS3.** Design and integrate intelligent network architectures by coordinating AI models, devices, data sources, and digital platforms.
- GS4.** Utilize cloud and edge computing collaboratively to optimize system latency, bandwidth usage, scalability, and overall performance.
- GS5.** Validate and evaluate integrated AI and digital solutions through systematic testing, performance monitoring, and compliance verification to ensure operational effectiveness.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	15	-	-
PC1. Monitor changes in risk profiles and client needs to determine appropriate maintenance, servicing, or upgrade actions for security systems.	-	2	-	-
PC2. Perform routine maintenance and servicing activities in accordance with established security industry maintenance standards and procedures.	-	2	-	-
PC3. Conduct software unit testing and ICT troubleshooting to identify faults and performance issues within security systems.	-	2	-	-
PC4. Test sub-systems and integrated components to verify correct operation or diagnose malfunctions accurately.	-	2	-	-
PC5. Inspect systems to identify problems such as loose or damaged connections, component degradation, or equipment failure.	-	2	-	-
PC6. Evaluate repair versus replacement options and carry out corrective actions to restore system functionality effectively.	-	1	-	-
PC7. Verify that existing security systems continue to meet current technical, safety, and regulatory standards.	-	1	-	-
PC8. Advise clients on equipment upgrades, regulatory changes, and functional improvements to maintain system effectiveness and client confidence.	-	1	-	-
PC9. Implement sustainable practices by recycling replaced equipment and materials in accordance with environmental guidelines.	-	1	-	-
PC10. Provide backup, disaster recovery support, and coordinate multi-system operational maintenance while completing and sharing required reports and documentation with relevant stakeholders.	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	-	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6723
NOS Name	Maintenance and repair
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

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5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on this criterion.

6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6716.Work organization and management (Intelligent Security Technology)	-	5	-	-	5	5
ELE/N6717.Communication and interpersonal skills (Intelligent Security Technology)	-	5	-	-	5	5
ELE/N6724.Project initiation and holistic planning. (Intelligent Security Technology)	-	5	-	-	5	5
ELE/N6720.Planning and design of security systems (Intelligent Security Technology)	-	20	-	-	20	20
ELE/N6719.Installation and implementation of security systems (Intelligent Security Technology)	-	10	-	-	10	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6721.Debugging and optimization of security systems (Intelligent Security Technology)	-	30	-	-	30	30
ELE/N6722.Application of artificial intelligence and digital technologies (Intelligent Security Technology)	-	10	-	-	10	10
ELE/N6723.Maintenance and repair	-	15	-	-	15	15
Total	0	100	-	-	100	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.